Epistemic Hedging is Linear in Pretrained Sentence Embeddings

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Abstract

Epistemic hedging — words like "probably," "certainly," and "possibly" — has been psychometrically calibrated since the 1960s, with median probabilities for verbal expressions remarkably stable across more than fifty years of replication. We investigate whether this calibration is recoverable as a **verbal probability axis** in frozen pretrained pooled sentence embeddings. Within syntactic types, verbal probability is the dominant linear direction across five architecturally diverse embedding models (supervised Spearman $\rho > 0.90$), and the axis triangulates against two independent psychometric datasets at $\rho \approx 0.97$ in both. The same axis transfers in ranking across eight typologically diverse languages via a multilingual encoder, is robust to an order of magnitude of dimensional compression on a Matryoshka-trained model, and passes permutation and label-shuffling null tests on the three larger-n syntactic types. Rank-1 representation-level erasure of the axis clears two complementary nulls — a cosine-matched random-direction control and a label-permutation null — at $p = 0.002$ on the headline cross-type pair across two architectures, the embedding-class structural analogue of activation-steering experiments standard in the decoder-LLM linear-feature literature. We frame the empirical convergence across three psychometric datasets, six embedding architectures, and eight languages as evidence that distributional embedding geometry and human probability semantics are convergent measurements of one underlying epistemic dimension.

1 Introduction

Mikolov et al. (2013) demonstrated that semantic relationships manifest as consistent vector directions in word embedding space — *king* $-$ *man* $+$ *woman* $\approx$ *queen*. The Linear Representation Hypothesis (LRH) generalizes this observation: many human-meaningful concepts correspond to linear directions in language model representations (Park et al., 2024). Three lines of recent work bear on whether epistemic content fits this pattern (§2): linear features in decoder LLM internal activations (Marks and Tegmark, 2023; Bürger et al., 2024; Yu et al., 2025; Ji et al., 2025; Lepori et al., 2026; Valois et al., 2025); verbal-probability semantics elicited behaviorally from language model outputs (Sileo and Moens, 2023; Belém et al., 2024; Tang et al., 2026; Wang et al., 2024); and probing of pooled sentence representations for graded epistemic content (Jacobs et al., 2022; Chen et al., 2020; Schuster et al., 2020; Pei and Jurgens, 2021). None demonstrates a *continuously calibrated verbal probability axis* in *frozen pretrained pooled sentence embeddings*, *aligned with human psychometric judgments*, and *validated against independent psychometric datasets*. We address this gap using the Mosteller and Youtz (1990) dataset of 53 verbal expressions rated by 238 science writers (e.g., "likely" $\rightarrow$ median 71.1%, "possible" $\rightarrow$ 38.5%), embedded in controlled templates, with difference vectors from bare claims trained into a probability axis and evaluated against two independent psychometric datasets (Vogel et al., 2022; Wintle et al., 2019), eight typologically diverse languages, dimensional truncation, null controls, and **concept erasure as functional validation** — the embedding-class analogue of decoder-LLM causal intervention.

Our contributions are: (1) a calibrated within-type verbal probability axis triangulated across three psychometric datasets ($\rho > 0.90$ supervised, 0.59–0.94 LOO; Mosteller$\rightarrow$Vogel $\rho = 0.991$, Mosteller$\rightarrow$Wintle $\rho = 0.967$); (2) an embedding-class structural analogue of decoder-LLM causal intervention via rank-1 concept erasure (predicative↔modal ΔMAE $= +17$ to $+25$pp clearing both a cosine-matched random-direction null and a label-permutation null at $p = 0.002$; cross-pair $r(\cos, \Delta\text{MAE}) = 0.92/0.86$); and (3) cross-architecture and cross-linguistic robustness across six models in five architectural families and eight typologically diverse languages (mean ranking $|\rho| = 0.93$), with three informative negatives (no IQR encoding, no consistent ensemble gain over modal-only, near-orthogonal negation as probability-reflection).

2 Related Work

Linear features in decoder LLM internal activations. The Linear Representation Hypothesis (Park et al., 2024) has been tested most extensively on decoder LLM internal states. Marks and Tegmark (2023) identify a truth direction in residual streams; Bürger et al. (2024) refine it for polarity; Yu et al. (2025) generalize one-dimensional truth directions to cones. The two most direct comparators are Ji et al. (2025), who locate a Verbal Uncertainty Feature in 7B-class LLM residual streams via difference-in-means with LLM-as-a-Judge labels and validate via activation steering, and Lepori et al. (2026), who extract modal difference vectors from late-layer hidden states discriminating expert-labeled modal categories. Valois et al. (2025) extend the LRH to multi-token frames. All operate on decoder LLM internal activations and validate via token-level causal intervention.

Verbal-probability semantics elicited behaviorally. A parallel literature interprets hedges through prompting rather than representation (Sileo and Moens, 2023; Belém et al., 2024; Tang et al., 2026; Wang et al., 2024); semantics are located in outputs, not geometrically in representations.

Pooled sentence representation probing. The closest precursor is Jacobs et al. (2022), who use mean-pooled RoBERTa with the critical word masked to predict cloze probability via per-item classifiers; their target is sentential constraint via next-word predictability, where ours is direct ridge regression of pooled-embedding differences onto psychometric medians for the explicit hedge lexicon. Chen et al. (2020) fine-tune BERT for NLI entailment probability; Schuster et al. (2020); Pei and Jurgens (2021); Lee et al. (2015); Stanovsky et al. (2017); Bhatia (2017); Shen et al. (2023) locate scalar implicature, rhetorical certainty, factuality, event likelihood, or embedding noise — none extract a frozen-geometry verbal-probability axis cross-validated against independent psychometric datasets.

The gap. No prior work demonstrates the conjunction this paper fills: a *continuously calibrated* verbal-probability axis, in *frozen pretrained pooled sentence embeddings*, *aligned with human psychometric judgments*, and *validated against independent psychometric datasets* spanning over fifty years (Mosteller and Youtz, 1990; Vogel et al., 2022; Wintle et al., 2019). The contribution is parallel in spirit to the decoder-LLM linear-feature literature, conducted in a model class where token-level causal intervention is unavailable and representation-level concept erasure (§4.4) is the appropriate analogue.

3 Method

We extract a verbal probability axis from frozen pretrained pooled sentence embeddings, calibrate it against human psychometric medians, and evaluate it against in-sample fit, leave-one-out generalization, two independent psychometric datasets, eight typologically diverse languages, several baselines, three null-hypothesis controls, and a representation-level concept-erasure protocol. The protocol is identical across experiments: every script in the codebase is self-contained and reuses the same difference-vector construction, ridge-regression objective, and linear calibration step, parameterized only by the syntactic type of the hedge expressions and the embedding model. Math notation: vectors are column vectors, $\|\cdot\|$ is the ℓ_2 norm, $\hat{v} = v/\|v\|$ denotes the unit-normalized form of v, ρ is Spearman rank correlation, and MAE is mean absolute error in percentage points.

3.1 Data

Primary calibration: Mosteller and Youtz (1990). We work from the Mosteller and Youtz (1990) corpus of 53 verbal probability expressions rated by 238 science writers, with 25th-percentile, median, 75th-percentile, and inter-quartile-range values on a $[0, 100]$ scale. One row ("Doubtful") lacks all percentile and median fields in the source and is therefore excluded from every analysis; the remaining 52 entries form the **Mosteller-derived analysis set** the paper draws on. We use the median as the target probability throughout; the IQR is reserved for the precision-encoding probe in §4.7.

Cross-validation datasets. Two independent psychometric datasets test whether the trained axis generalizes beyond the calibration source. Vogel et al. (2022) is a systematic review and meta-analysis of 21 verbal-probability studies spanning 1967–2018; we extract the median pooled across studies for each expression that matches our syntactic types. Wintle et al. (2019) ($n \approx 924$) provides a third independent dataset; we transcribe nine predicative and three frequency-adverbial values from their Figure 2 and Table 1. All three sources collect aggregate human probability judgments under non-verifiable contexts following the methodological framework of Wallsten et al. (1986) and Budescu and Wallsten (1995); the convergence of three datasets collected by different groups in different decades on different populations is the basis for the cross-dataset generalization claim in §4.2.

Type classification. Mixing syntactic types confounds the analysis: PCA on the 52 mixed-type Mosteller-derived difference vectors (one sentence per source entry with a median) recovers PC1 as a syntactic-frame direction (28.7% of variance, ordering noun-phrase constructions against frequency adverbs) rather than as the probability axis (§4.1). Within-type analysis is therefore not a methodological convenience but a load-bearing decision that isolates the semantic signal from the syntactic envelope (§4.1; cf. the polarity confound Bürger et al., 2024 report for truth directions). We partition the analysis set into four syntactic types, each with a single sentence template that holds syntax constant across all expressions of that type. With s_0 denoting the bare claim "The experiment will succeed," the four types and their templates (with hedge expression x substituted in) are:

- **Predicative adjective** ($n = 13$): "It is x that the experiment will succeed."
- **Frequency adverb** ($n = 19$): "The experiment will x succeed."
- **Noun phrase** ($n = 11$): "There is a(n) x that the experiment will succeed."
- **Modal adverb** ($n = 10$): "The experiment will x succeed."

The frequency-adverb and modal-adverb templates share a syntactic slot but encode different semantic types (base-rate frequency versus epistemic confidence); the modal axis is not derivable from the frequency axis (§4.1). The modal-adverb items are derived as adverbial forms of corresponding predicative entries (e.g., the modal "certainly" reuses the median of the predicative "Certain"), so the four within-type partitions sum to 53 template-instance items distributed across 43 unique source rows.

Three modal subsets appear in the codebase, and we name them explicitly to keep claims precise. **M10** is the ten Mosteller-grounded adverbial forms (`certainly`, `almost certainly`, `very likely`, `likely`, `probably`, `very probably`, `possibly`, `unlikely`, `very unlikely`, `improbably`); it is the canonical modal axis used in §4.1 (Table 1), §4.4 (concept erasure), and §4.6 (null-hypothesis controls). **M15** is an expanded set of fifteen items — eight of M10 plus seven author-estimated additions (`conceivably`, `definitely`, `perhaps`, `maybe`, `presumably`, `undoubtedly`, `arguably`) — used for sample-size headroom in the §4.5 robustness probes (truncation, compound composition, bimodality) and §4.7 (ensemble baseline). M15 trades calibration grounding for size, and §4.5 prose marks results that depend on it as exploratory rather than calibrated. A third translation-pairing modal subset (10 items, of which 7 are M10-derived and 3 are author-estimated) is used by the §4.3 cross-linguistic experiment for translation parity across languages and is treated similarly.

3.2 Models

We test six frozen pretrained embedding models spanning five architectural families and dimensions from 768 to 4096: nomic-embed-text v1.5 (768d, BERT + Matryoshka; default for the within-type analyses), embeddinggemma 300M (768d, decoder-derived from Gemma), nomic-embed-text v2 MoE (768d, Mixture of Experts), mxbai-embed-large (1024d, BERT-large), qwen3-embedding (4096d, Qwen3 architecture), and bge-m3 (1024d, XLM-RoBERTa-based, multilingual; used for the §4.3 cross-linguistic transfer). We make no fine-tuning, no head training, and no architectural modifications; the qualifier "frozen pretrained" in the novelty claim depends on this.

The five within-type models span five distinct architectural families (BERT, BERT-large, Mixture of Experts, decoder-derived heads, and Qwen3) and three training paradigms; bge-m3 adds a multilingual XLM-R-based encoder trained on more than one hundred languages. All five within-type models output a single mean- or [CLS]-pooled sentence vector per input. Cross-architecture robustness (§4.1) is the basis for the claim that the verbal probability axis is a property of pretrained pooled-embedding geometry rather than of any specific architecture or training corpus.

3.3 Axis extraction

Difference-vector construction. For each syntactic type T with expressions $\{x_1, \ldots, x_n\}$ and Mosteller medians $\{y_1, \ldots, y_n\}$, we embed the bare claim s_0 and each templated sentence $T(x_i)$ using the model's pooled output, yielding $\mathbf{e}_0$ and $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ in $\mathbb{R}^d$. The input feature for expression x_i is the difference vector

$$\mathbf{d}_i \;=\; \mathbf{e}_i - \mathbf{e}_0,$$

which subtracts off the propositional content shared across all expressions in the type and isolates the contribution of the hedge phrase. Differencing against a shared bare claim is the same construction Mikolov et al. (2013) used to expose word-level analogies and that the decoder-LLM linear-feature literature uses to extract truth and uncertainty directions from residual streams (Marks and Tegmark, 2023; Ji et al., 2025).

Ridge regression on Mosteller medians. We standardize the medians $\tilde{y}_i = (y_i - \bar{y})/\mathrm{std}(y)$ and stack the difference vectors as rows of $D \in \mathbb{R}^{n\times d}$. The probability axis $\hat{\mathbf{w}}$ is the unit-normalization of the ridge regression (Hoerl and Kennard, 1970) solution with regularization $\lambda = 0.1$:

$$\mathbf{w}^* \;=\; \left(D^\top D + \lambda I_d\right)^{-1} D^\top \tilde{\mathbf{y}}.$$

The system is severely under-determined ($d \in \{768, 1024, 4096\}$, $n \leq 19$); the ridge term selects the minimum-norm solution along the dominant probability-correlated direction. $\lambda = 0.1$ is fixed across all experiments for cross-cell comparability; a four-decade sweep over $\lambda \in \{0.001, \ldots, 10\}$ on two models $\times$ four types confirms LOO ρ stays within 0.072 of the reference everywhere, with the modal axis most exposed to regularizer choice (cf. §6 L4).

Linear calibration. We recover percentage-scale predictions via OLS of in-sample projections $p_i = \mathbf{d}_i^\top \hat{\mathbf{w}}$ against unstandardized medians y_i, fitting (α, β) such that $\hat{y}_i \;=\; \alpha p_i + \beta$, clipped to $[0, 100]$. We train one $(\hat{\mathbf{w}}, \alpha, \beta)$ triple per (model, type) pair. Cross-content validation on alternative bare claims ("the treatment will be effective", "the prediction will be correct") preserves axis ranking with $\rho > 0.69$.

3.4 Evaluation strands

We evaluate along six strands: (1) **in-sample fit and leave-one-out** with bootstrap 95% CIs (§4.1); (2) **cross-dataset evaluation**, applying the Mosteller-trained axis without retraining to Vogel et al. (2022) and Wintle et al. (2019) as population-level psychometric replication (§4.2); (3) **unsupervised baselines** — random direction, mean-difference centroid (Marks and Tegmark, 2023; Ji et al., 2025), PCA PC1 (§4.1); (4) **null-hypothesis controls** — label permutation, evaluative non-epistemic adjectives, Uniform$(0, 100)$ labels (§4.6); (5) **cross-linguistic transfer** via bge-m3 to eight languages, with Spearman ρ load-bearing and MAE diagnostic (§4.3, §6 L2); (6) **concept erasure** by rank-1 projection $\mathbf{d}_i' = \mathbf{d}_i - (\mathbf{d}_i^\top \hat{\mathbf{w}}_A)\hat{\mathbf{w}}_A$ followed by re-decoding with type B's calibrated triple, evaluated against cosine-matched random-direction and label-permutation nulls (§4.4).

4 Results

4.1 The syntactic confound and within-type probability axes

When the four syntactic types of §3.1 are pooled, the dominant linear direction in embedding space is *not* probability but syntax. PCA on the 52 mixed-type difference vectors from nomic-embed-text v1.5 returns PC1 explaining 28.7% of the variance and ordering constructions by template family — noun-phrase forms ("high probability") at one end, frequency adverbs ("very infrequent") at the other; probability appears only on PC2 ($\rho = -0.68$, $p = 2.4 \times 10^{-8}$), and PC1 correlates only weakly with the medians ($\rho = -0.28$). This is the structural counterpart of the polarity confound Bürger et al. (2024) report for truth directions in decoder LLM residual streams.

Holding type fixed reverses the picture. Within each type partition, a ridge axis $\hat{\mathbf{w}}$ trained on Mosteller medians (§3.3) recovers the probability ordering with high in-sample fidelity and generalizes through leave-one-out (LOO). Table 1 reports both across five models: in-sample Spearman ρ exceeds 0.90 for every (model, type) cell, and LOO ρ falls in 0.59–0.94. The two cells below 0.65 (modal at $n = 10$) are the small-sample

Model	Predicative (n=13)	Frequency adverb (n=19)	Noun phrase (n=11)	Modal adverb (n=10)
	sup. / LOO	sup. / LOO	sup. / LOO	sup. / LOO
nomic-embed-text v1.5	0.934 / 0.874	0.949 / 0.923	0.973 / 0.936	0.927 / 0.648
embeddinggemma-300M	0.989 / 0.874	0.951 / 0.849	0.955 / 0.836	0.939 / 0.588
nomic v2 MoE	0.945 / 0.819	0.963 / 0.869	0.936 / 0.855	0.952 / 0.915
mxbai-embed-large	0.967 / 0.912	0.946 / 0.905	0.973 / 0.855	0.976 / 0.806
qwen3-embedding	0.901 / 0.747	0.916 / 0.704	0.964 / 0.773	0.952 / 0.855

ρ-saturation regime (§6 L4). Probability dominates the within-type cloud without supervision as well: PCA PC1 within type reaches $\rho = 0.98$ on nomic noun phrases (PC1 = 36% variance), 0.93 on mxbai predicative (PC1 = 60%), 0.96 on qwen3 noun phrases (PC1 = 61%).

Table 1. Within-type Spearman ρ between calibrated predictions and Mosteller medians, in-sample (sup.) and leave-one-out (LOO), across five frozen pretrained pooled sentence embedding models. All four type-axes are trained and evaluated on the Mosteller-grounded subsets defined in §3.1 (modal axis: ten adverbial expressions for which Mosteller and Youtz (1990) report a median; the seven additional author-estimated modal terms used in early exploratory work are excluded).

The pattern holds across five architectural families (BERT, BERT-large, MoE, decoder-derived heads). The weakest LOO cells track $n = 10$ (modal) rather than any model-specific failure mode (§6 L4). Figure 1 contrasts the mixed-type PCA (PC1 = syntax, PC2 = probability) against the predicative-only PCA (PC1 alone orders monotonically "impossible" $\rightarrow$ "certain").

4.2 Three-dataset psychometric triangulation

The Mosteller and Youtz (1990) calibration is sixty years old and has been replicated repeatedly under different protocols and on different populations. We test whether the within-type axes trained on Mosteller medians transfer, *without retraining*, to two independently collected psychometric corpora: the Vogel et al. (2022) systematic review and meta-analysis aggregating 21 verbal-probability studies from 1967 to 2018, and the Wintle et al. (2019) experiment ($n \approx 924$) drawing on a different population with a different elicitation protocol. The Mosteller-trained axis $(\hat{\mathbf{w}}, \alpha, \beta)$ is applied as in §3.4 to surface-form-matched expressions embedded in the same syntactic templates; we report Spearman ρ and MAE on the Vogel and Wintle medians.

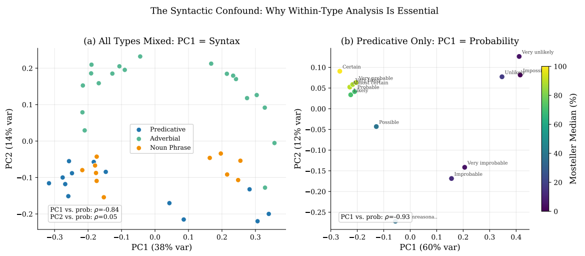

Figure 1: Mixed-type versus within-type PCA on the Mosteller difference vectors (nomic-embed-text v1.5). (a) Mixed-type PC1 (28.7% variance) separates by syntactic frame, not by probability ($\rho = -0.28$); the probability signal is on PC2 ($\rho = -0.68$). (b) Within-type predicative-only PC1 orders expressions monotonically by Mosteller median.

For predicative expressions — the type with the largest cross-corpus overlap ($n_{\text{test}} = 11$ for Vogel, 9 for Wintle) — Mosteller-to-Vogel correlations exceed 0.89 across all five models and reach 0.991 on mxbai-embed-large with MAE = 5.1 percentage points (Table 2). Mosteller-to-Wintle predicative results are comparable: $\rho = 0.967$ on mxbai-embed-large (MAE = 3.6%) and $\rho \geq 0.917$ on nomic-embed-text v1.5 and qwen3-embedding (MAE between 3.5% and 6.9%). The frequency-adverb cross-validation is more limited by overlap ($n_{\text{test}} = 3$ on both Vogel and Wintle), but Vogel returns $\rho = 1.000$ on three of the five models with MAEs between 2.4% and 6.7%, and Wintle returns $\rho = 1.000$ on every tested model with MAEs of 4.0–10.5%. Noun-phrase expressions have no cross-corpus matches and are not included in the table; the cross-corpus tests therefore probe predicative and frequency-adverbial calibration directly.

Table 2. Cross-dataset transfer for the predicative axis. The Mosteller-trained axis is applied

Model	Mosteller→Vogel predicative (n=11)	Mosteller→Wintle predicative (n=9)
	ρ / MAE	ρ / MAE
nomic-embed-text v1.5	0.946 / 7.1%	0.917 / 6.9%
embeddinggemma 300M	0.964 / 5.6%	—
nomic v2 MoE	0.891 / 6.5%	—
mxbai-embed-large	**0.991 / 5.1%**	**0.967 / 3.6%**
qwen3-embedding	0.909 / 5.6%	0.917 / **3.5%**

without retraining to Vogel et al. (2022) and Wintle et al. (2019) median probabilities. Frequency-adverbial cross-validation ($n_{\text{test}} = 3$ in both cases) is reported alongside.

Three observations follow. First, mxbai's $\rho = 0.991$ against Vogel is indistinguishable from its in-sample Mosteller fit (0.967), indicating the axis recovers a population-level mapping rather than overfitting Mosteller's 1990 sample. Second, convergence is architecture-independent: BERT-base, BERT-large, MoE, and decoder-derived heads all clear $\rho \geq 0.89$. Third, Wintle's calibration is recovered with MAE as low as 3.5–3.6% — tighter than in-sample Mosteller LOO MAE — consistent with Wintle's larger pool ($n \approx 924$) producing tighter population medians. The three sources were collected by different groups in different decades on different populations using different protocols; their joint agreement with the embedding-derived axis argues against any single-corpus artifact.

4.3 Cross-linguistic ranking transfer

We test whether the verbal probability axis is a property of English specifically or of multilingual pretrained representations more generally. Using bge-m3 (1024d, XLM-RoBERTa-based, multilingual; Chen et al., 2024), we train predicative and modal axes on a translation-paired English subset and apply each axis without retraining to translated hedge expressions in eight typologically diverse languages (German, French, Spanish, Mandarin Chinese, Japanese, Korean, Arabic, Hindi). The English training subsets are translation-paired

Language	Predicative ρ (n=8)	Predicative MAE	Modal ρ (n=6–7)	Modal MAE	Predicative axis cos vs. EN
German	0.952	8.3%	0.918	7.1%	0.940
French	0.952	7.3%	0.946	8.4%	0.941
Spanish	0.905	9.7%	0.852	9.1%	0.904
Chinese	0.952	12.4%	**0.991**	10.1%	0.779
Japanese	0.952	10.4%	0.857	**18.1%**	0.845
Korean	**1.000**	8.3%	0.841	8.2%	0.865
Arabic	0.833	6.9%	0.943	**3.6%**	0.859
Hindi	0.952	12.5%	**1.000**	6.5%	0.845

rather than the canonical sets of §3.1: a 12-item predicative subset (drops "Not unreasonable", which lacks a clean per-language translation) and a 10-item modal subset of which seven are M10-derived and three are author-estimated terms ("definitely", "perhaps", "undoubtedly") retained for cross-language parity. The expected probability of each translated expression is the Mosteller median of its English equivalent — a translation pairing, not a native-speaker psychometric value (§6 L2). Spearman ρ between ranked projections and the translation-paired targets is therefore the load-bearing **ranking-transfer** metric; MAE is reported as a translation-pairing diagnostic only.

Table 3. Zero-shot ranking transfer of the English-trained bge-m3 axis to eight typologically diverse languages. Predicative axis cos compares the English-trained axis to a per-language axis trained on translated Mosteller equivalents (§3.3 protocol applied per language). Mean $|\rho|$ across all 16 (language, frame) cells = 0.928.

Sixteen of sixteen (language, frame) combinations clear $\rho \geq 0.83$, with mean $|\rho| = 0.928$. Korean predicative ($\rho = 1.000$, $n = 8$), Hindi modal ($\rho = 1.000$, $n = 6$), and Chinese modal ($\rho = 0.991$, $n = 7$) span typologically distant Indo-European, Indo-Aryan, and Sino-Tibetan branches. Per-language axis-alignment cosines (Table 3, last column) — training a separate ridge axis on each language's translated equivalents and

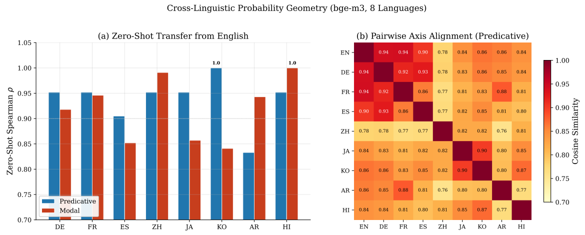

Figure 2: Cross-linguistic ranking transfer (bge-m3, English-trained axis applied zero-shot to eight languages). Per-language projection of translated predicative expressions onto the English-trained axis; rank order is preserved across all eight languages, with Korean at $\rho = 1.000$.

measuring against the English-trained axis — cluster at 0.94 for German, French, and Spanish and at 0.85–0.87 for Korean, Japanese, Arabic, and Hindi; Japanese and Korean are the only pair whose predicative axes align more strongly with each other (cos = 0.900) than with English, consistent with their parallel epistemic-marker morphology.

The MAE column tells a more nuanced story. Arabic modal MAE reaches 3.6% — comparable to Wintle's. Japanese modal MAE is 18.1%, driven by translations whose pairing is a known approximation: pragmatic-loaded Japanese epistemic markers carry contextual implicatures a template-based evaluation cannot capture, and the "expected" probability column is itself a translation-pairing artifact rather than a native-speaker calibration. The ranking ($\rho = 0.857$ on Japanese modal) is preserved; only the calibration MAE diverges. §6 L2 develops the limits of this protocol in full.

4.4 Functional validation via concept erasure

Rank-1 erasure of axis $\hat{\mathbf{w}}_A$ from type-B difference vectors followed by re-decoding with type B's calibrated triple, applied across all twelve ordered pairs (A, B) with $A, B \in$ {predicative, frequency adverb, noun phrase, modal} on mxbai-embed-large and qwen3-embedding, yields a cross-model structural signature: the Pearson correlation between inter-axis cosine $\cos(\hat{\mathbf{w}}_A, \hat{\mathbf{w}}_B)$ and the calibrated ΔMAE after erasure is $r = 0.92$ (mxbai) and 0.86 (qwen3) across all twelve pairs (Figure 3). Erasing a probability-aligned axis degrades the downstream decoder in proportion to the probability content shared with the target, and this scaling holds across a 1024d BERT-large variant and a 4096d decoder-derived embedding head. We treat this cross-pair signature, rather than per-pair verdicts, as the load-bearing finding: the four within-type axes are partially redundant projections of a shared epistemic dimension (§5).

The predicative–modal pair has the highest cosine alignment in both models (0.88 mxbai, 0.79 qwen3). Erasing predicative from modal raises MAE from 6.8 to 25.2 on mxbai (+18.4) and 5.2 to 22.5 on qwen3 (+17.4); the reverse raises 4.6 to 29.2 on mxbai (+24.6) and 5.5 to 26.0 on qwen3 (+20.5). We test against two complementary nulls: a cosine-matched random-direction control (does this *direction* damage more than a random direction at the same geometric overlap?) and a label-permutation null (does the eraser's *probability content* do the work, retraining on shuffled medians?). The headline pair clears both. Real-axis ΔMAE exceeds matched-random with ratios 1.02–1.23 across the four model × direction combinations (matched-random is stringent at cos = 0.88, reproducing ~98% of MAE damage geometrically; §6 L3). Against label-permutation, real ΔMAE exceeds every one of 500 shuffles, $p = 0.002$ in all four cases (e.g., mxbai Pred→Modal real $+18.4$ vs. null $+1.2 \pm 1.7$; qwen3 Modal→Pred $+20.5$ vs. $+2.7 \pm 3.9$).

We use MAE rather than ρ because in-sample ρ already ≥ 0.95 leaves no headroom (§6 L4): the ρ-view returns 5/12 functional on mxbai and 3/12 on qwen3 with zero overlap, while $r(\cos, \Delta\text{MAE}) = 0.92/0.86$ contrasts with $r(\cos, \Delta\rho) = -0.72/+0.24$. On qwen3, eleven of twelve pairs clear both nulls; on mxbai, five clear both, four clear matched-random only. Zero pairs in either model clear label-permutation only — where the nulls disagree, label-permutation is more conservative. The result reproduces under leave-one-out target decoding (refitting $(\hat{\mathbf{w}}_B, \alpha_B, \beta_B)$ on $n_B - 1$ items per held-out target), with cross-pair r rising to $+0.98$ (mxbai) and $+0.89$ (qwen3) and the headline pair clearing the matched-random LOO null in all four combinations ($z = 3.7$–12.8). This is the embedding-class structural analogue of activation-steering protocols in the decoder-LLM linear-feature literature (Ji et al., 2025; Marks and Tegmark, 2023; Valois et al., 2025; Lepori et al., 2026); the "analogue rather than substitute" caveat appears in §6 L3.

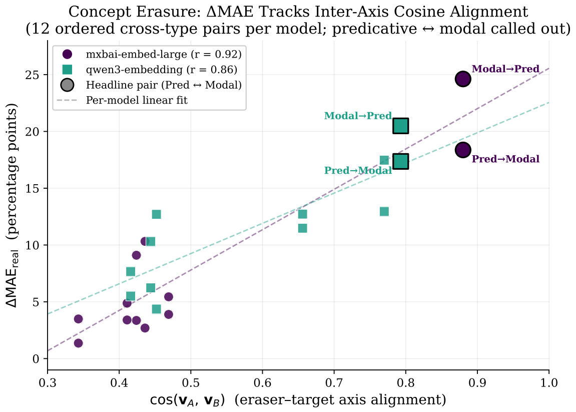

Figure 3: ΔMAE vs. cosine alignment between erased and target axes, across 12 ordered cross-type pairs per model. mxbai-embed-large (filled circles) Pearson $r = 0.92$; qwen3-embedding (filled squares) $r = 0.86$. Headline pair (predicative ↔ modal, both directions, both models) called out at larger marker size with bold annotation. The cross-pair $r \approx 0.9$ across both architectures is the cross-model-robust empirical signature of the §4.4 functional-validation claim.

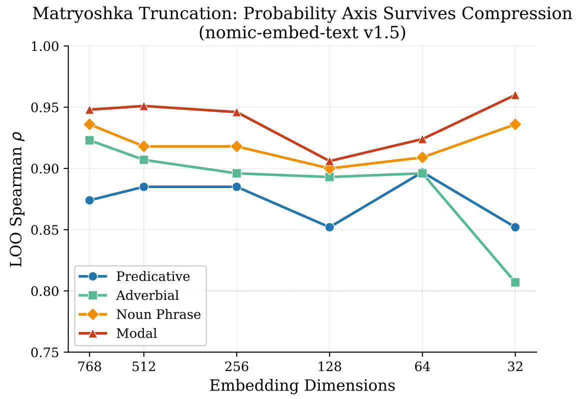

Figure 4: Matryoshka dimensional truncation: LOO ρ versus truncated embedding dimension on nomic-embed-text v1.5 (MRL-trained, 768d) and mxbai-embed-large (non-MRL, 1024d). The MRL-trained model preserves the axis to 64 dimensions (12× compression); the non-MRL model holds at 128 dimensions.

4.5 Robustness: dimensional truncation, novel-phrase generalization, compound composition, bimodality probe

Four robustness probes test the axis along directions orthogonal to the Mosteller-Vogel-Wintle calibration.

Matryoshka dimensional truncation. Following Kusupati et al. (2022), we re-extract within-type axes after truncating pooled embeddings to 512, 256, 128, 64, 32 dimensions. On nomic-embed-text v1.5 (768d, MRL-trained), the worst-case LOO ρ drop at 64d across predicative, frequency-adverb, and noun-phrase axes is 0.027 — a 12× compression preserving rank recovery. The canonical M10 modal axis is more delicate: full-dimensional nomic modal LOO ρ is 0.648 because "improbably" often functions as an intensifier ("improbably handsome") rather than a hedge, embedding closer to 50% than its 12.5% source-adjective median. Truncation mitigates this distortion (modal LOO rises to 0.806 at 64d). On mxbai-embed-large (non-MRL, 1024d), the three larger axes hold to 128d (8×) but degrade at 64d (drop 0.154), as expected for a loss without nested-prefix enforcement. Axis-stability cosines confirm the dichotomy: nomic retains cos ≥ 0.94 at 64d, mxbai ≥ 0.92 at 128d falling to 0.85–0.93 at 32d.

Novel-phrase generalization, compound composition, and bimodal disambiguation. (i) Applying trained axes to 26 phrases outside Mosteller's vocabulary (informal hedges, doubt, anti-hedges, "signs point to") against author-rated probabilities yields $\rho = 0.75$–0.94 across five models (0.80–0.94 on the ten purely novel phrases); the geometry orders intuitively (believe > think > imagine > suspect > wonder > doubt) with center-compression from natural-sentence dimensional distribution — evidence of generalization, not calibrated prediction (§6 L1). (ii) Compound difference vectors $\mathbf{d}_{A+B}$ align near-linearly with $\mathbf{d}_A + \mathbf{d}_B$ on qwen3 and mxbai (cos 0.82–0.89 across first-person + modal, evidential + modal, attenuator + predicative, double first-person), with ≈0.75 magnitude attenuation — sub-linear in magnitude, near-linear in direction. Negation is structurally exceptional: $\mathbf{d}_{\neg A}$ is near-orthogonal to $\mathbf{d}_A$ (cos 0.22 on qwen3 for "not impossible" vs. "impossible") and acts as a probability-reflection (+0.96 between negations of high-probability words, −0.93 between negations of low/high pairs). (iii) "{modifier} possible" expressions traverse 20–32 pp on the predicative axis between low-leaning ("barely possible") and high-leaning ("eminently possible") compositions across three models, reaching the bimodal range of human "possible" interpretations (5–10% vs. 40–50%) even though bare-"possible" projects to a single point.

4.6 Null-hypothesis controls

Three controls on nomic-embed-text v1.5 establish that the within-type LOO recovery of §4.1 is in the phrase-to-probability mapping rather than in the protocol or template structure.

Permutation test. Permuting Mosteller medians within type and rerunning the ridge-plus-LOO pipeline 1{,}000 times yields permuted-ρ distributions centered near zero (σ 0.29–0.42). Real LOO clears the 99th percentile on the three larger-n axes (predicative $\rho = 0.874$, $p = 0.003$; frequency adverb 0.923, $p < 0.001$; noun phrase 0.936, $p = 0.005$) and passes on three additional architectures at $p \leq 0.029$ across nine cells. The modal axis at $n = 10$ shows a cross-model gradient tracking LOO ρ: embeddinggemma (0.59, $p = 0.22$) and nomic v1.5 (0.65, $p = 0.21$) fail; mxbai (0.81, $p = 0.054$) borderline; nomic v2 MoE (0.92, $p = 0.004$) passes. A multi-n power analysis appending author-estimated modal items at $n \in \{10, 12, 14, 17\}$ collapses p three orders of magnitude over $n = 10 \rightarrow 14$, crossing $p <$ 0.05 between 10 and 12 (probes sample-size variation, not calibration). §6 L4 develops the small-n reading; the modal axis is independently validated functionally in §4.4 at $n = 10$.

Evaluative non-epistemic control. Thirteen *that*-complement adjectives spanning natural valence ("tragic" through "wonderful") in the same predicative template, trained against scrambled labels, return LOO $\rho = -0.41$ — far below the epistemic 0.874 and on the opposite side of zero. The negative sign is interpretable: ridge detects the embedding's real evaluative axis, so LOO predictions track natural valence rather than scrambled labels, consistent with the multiple-axis picture §4.1 establishes. A *mixed* set including epistemic-leaky adjectives ("expected", "obvious") returns LOO $\rho = 0.69$ — evidence *for* the geometry's epistemic sensitivity (§6 L1).

Random-label control. Uniform$(0, 100)$ labels, 1{,}000 trials per type, match real-LOO in $\leq 6/1000$ on the three larger-n axes ($p \leq 0.006$); the modal axis at $n = 10$ matches in 197/1000, the same small-n low-power signature.

4.7 Negative results

Two informative negatives constrain the interpretation of the verbal probability axis: the IQR result tells us what kind of object the axis is, and the ensemble result tells us that the four within-type axes are partially redundant projections rather than independent measurements of a shared latent quantity.

Interpretive precision is not in the geometry. Mosteller and Youtz (1990) report inter-quartile range (IQR) per expression; "possible" has IQR = 42.7 (P25 = 7.5%, P75 = 50.2%; the bimodal distribution of §4.5), "certain" has IQR = 1.1. We test whether geometry encodes interpretive precision by correlating each expression's cross-claim consistency (mean pairwise cosine across 15 propositional contents) against $-$IQR. Across four models, $\rho(\text{consistency}, -\text{IQR})$ ranges from -0.20 to $+0.05$ ($p > 0.5$). The geometry encodes a single stable meaning per expression, not the spread of human interpretations. The axis is a *point estimator* on the population mapping, not a representation of its variance; this is consistent with treating hedges as continuous-valued operators on a single probability dimension (Lassiter, 2017), not as distributional objects encoding rater variance.

Modal axis as a strong but not strictly optimal single-axis baseline. Given that the four within-type axes are moderately correlated (cos = 0.41–0.88 on mxbai), we tested ridge ensemble, magnitude-weighted average, max-projection routing, logistic stacking, and naïve mean against M10 modal-only on the §4.5 novel-phrase set. Improvement is model-dependent: nomic v1.5 ensemble cuts MAE from 18.4% to 13.4%; qwen3 narrowly improves (7.7% vs. 8.7%); mxbai is tied (12.5% vs. 12.4%). The modal axis remains a strong single-axis fallback, consistent with the four axes being partially redundant projections of one shared epistemic dimension rather than independent measurements.

5 Discussion

The empirical content of this paper reads as evidence that distributional embedding geometry and human probability semantics are convergent measurements of one underlying quantity. Three psychometric datasets spanning fifty years (Mosteller and Youtz, 1990; Vogel et al., 2022; Wintle et al., 2019), six pretrained embedding architectures, and eight typologically diverse languages all agree on the same continuous probability axis. Each instrument carries different measurement noise — Mosteller's 238-rater sample, Vogel's 21-study meta-analytic aggregation, Wintle's $n \approx 924$ expert elicitation, and the architectural/training-data heterogeneity of the models — and their joint agreement on a common ordering and calibration is what one would expect if each were a noisy measurement of a single latent quantity. Schockaert (2022)'s formal framework for embeddings as epistemic states supplies the measurement-theoretic scaffolding. The §4.7 IQR negative sharpens the picture: the axis is a *point estimator* on the population mapping, recovering central tendency without dispersion, consistent with treating hedges as continuous-valued operators on a single probability dimension (Lassiter, 2017).

6 Limitations

We name four limitations the present study does not resolve. Remediation is framed as scope-of-future-work rather than as a flaw in the present claims.

(L1) Template-only evaluation; author-rated novel phrases. All §4.1–§4.4 results use controlled templates, so external validity to natural text is not rigorously established here. §4.5 partially mitigates: trained axes rank 26 non-template phrases at $\rho = 0.75$–0.94 against author-rated intuition; §4.6 shows the protocol does not trivially fit arbitrary labels (LOO $\rho = -0.41$ on scrambled valence labels). Real-text corpus calibration and a Prolific re-rating study on the §4.5 novel phrases are the natural follow-ups.

(L2) Cross-linguistic results are ranking transfer, not native-speaker calibration. The eight-language values used in §4.3 are translation pairings of English Mosteller equivalents rather than native-speaker psychometric surveys. Ranking ρ is the defensible claim; per-language MAE is a translation-pairing diagnostic only, especially for pragmatic-loaded markers (Japanese modal MAE = 18.1%). Native-speaker replication of Mosteller's protocol per target language is the remediation.

(L3) Non-causal embedding-class intervention. §4.4 establishes the axis is functional in the linear-decoder sense — the headline predicative↔modal pair clears both nulls on both architectures, with cross-pair $r(\cos, \Delta\text{MAE}) = 0.92/0.86$. This is the embedding-class analogue of activation-steering experiments (Ji et al., 2025; Lepori et al., 2026; Marks and Tegmark, 2023; Valois et al., 2025); pretrained pooled sentence embedding models lack a token-generation pathway, so representation-level erasure validated by a calibrated downstream linear decoder is the appropriate analogue rather than a substitute. The matched-random control is stringent at high cosine (at $\cos = 0.88$ it reproduces ~98% of MAE damage geometrically); the headline pair clears both it and the wider-variance label-permutation null in all four combinations.

(L4) Small per-type sample sizes saturate ρ-based metrics; the modal axis is most exposed. Within-type training uses $n = 10$ (modal), 11 (noun phrase), 13 (predicative), 19 (frequency adverb); in-sample $\rho \geq 0.95$ leaves no headroom for ρ-based erasure detection, which is why §4.4 leads with MAE. The same regime is visible in the §4.6 modal-axis permutation test: p tracks modal LOO ρ from failures at $\rho \approx 0.6$ through threshold at 0.81 to clean pass at 0.92, with the multi-n power-analysis curve crossing $p < 0.05$ between $n = 10$ and $n = 12$. This is a property of the available calibration data, not the extraction method.

7 Conclusion

Pretrained pooled sentence embedding models encode a continuously calibrated verbal probability axis aligned with human psychometric data: a single linear direction recovered within syntactic types, calibrated against Mosteller and Youtz (1990), cross-validated against Vogel et al. (2022) and Wintle et al. (2019), transferred zero-shot in ranking to eight typologically diverse languages, robust to dimensional truncation, and confirmed functional under two complementary erasure nulls. The result holds across six models spanning five architectural families, establishing for the pretrained pooled sentence embedding class a phenomenon the linear-feature interpretability literature has documented for decoder LLM internal states. Natural follow-ups: real-text corpus evaluation, native-speaker psychometric replication in the target languages, bridging embedding-class probes to decoder-LLM internal-state probes under shared input (Ji et al., 2025), and enlarging the modal psychometric corpus to alleviate the §6 L4 small-sample regime. Distributional embedding geometry and human psychometric surveys are most usefully read as convergent instruments measuring one underlying quantity.

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